10Pearls

Internship Project Report

AQI Predictor
Air Quality Index Prediction System

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January 2026 - February 2026

February 20, 2026

Abstract

This report presents the development of an automated machine learning system for predicting Air Quality Index (AQI) in Islamabad, Pakistan. The project implements a complete end-to-end pipeline including hourly data collection from OpenMeteo API, feature engineering, model training, and 72-hour forecast generation.

After evaluating multiple data sources (AQICN, OpenWeather), OpenMeteo was selected for its accessibility and comprehensive historical data. The system trains three gradient boosting models daily (Random Forest, XGBoost, LightGBM), with LightGBM achieving the best performance ($R^2 = 0.8229$, MAE = 14.68, RMSE = 34.86).

The production system uses MongoDB Atlas for data storage, GitHub Actions for automation, and Streamlit for visualization. All pipelines are fully automated, collecting data hourly and generating forecasts daily.

Keywords: Air Quality Index, Machine Learning, LightGBM, MongoDB, Automation, Pakistan

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Chapter 1

Introduction

1.1 Background

Air pollution is a critical environmental challenge in Pakistan. Accurate AQI prediction enables citizens to make informed decisions about outdoor activities and helps policymakers implement timely interventions. This project develops an automated machine learning system to forecast AQI values for Islamabad up to 72 hours in advance.

1.2 Objectives

1. Build a machine learning model to predict AQI with $R^2 \geq 0.75$
2. Implement automated hourly data collection
3. Create a production pipeline for daily model training and prediction
4. Deploy an interactive dashboard for visualization

1.3 Scope

Location: Islamabad, Pakistan (33.6996°N, 73.0362°E)

Included:

- Historical data collection (8,761 hourly records)
- AQI calculation using EPA standards
- Feature engineering (19 features)
- Three ML models (Random Forest, XGBoost, LightGBM)
- Automated deployment (GitHub Actions)
- MongoDB integration and Streamlit dashboard

Excluded:

- Real-time sensor integration
- Multi-city prediction
- Mobile application

Chapter 2

Methodology

2.1 Data Collection

2.1.1 Data Source Selection

Three APIs were evaluated:

Table 2.1: API Comparison

API	Access	Historical Data	Selected
AQICN	DNS blocked, VPN needed	2 months	No
OpenWeather	DNS blocked, VPN needed	2 months	No
OpenMeteo	No restrictions	60+ days	**Yes**

OpenMeteo was selected for: free access, no API key requirement, no DNS blocking, and sufficient historical data.

2.1.2 Data Variables

Air Quality (6 pollutants): PM10, PM2.5, CO, NO, SO, O

Weather (6 variables): Temperature, Humidity, Pressure, Wind Speed, Wind Direction, Precipitation

2.2 AQI Calculation

AQI is calculated using EPA's piecewise linear function:

$$AQI = \frac{I_{high} - I_{low}}{C_{high} - C_{low}} \times (C - C_{low}) + I_{low} \tag{2.1}$$

Overall AQI is the maximum value across all pollutants:

$$AQI_{overall} = \max(AQI_{PM2.5}, AQI_{PM10}, AQI_{O_3}, AQI_{NO_2}, AQI_{SO_2}, AQI_{CO}) \tag{2.2}$$

2.3 Feature Engineering

19 features were engineered across three categories:

Table 2.2: Feature Categories

Category	**Features**
Pollutants (6)	pm10, pm2_5, carbon_monoxide, nitrogen_dioxide, sulphur_dioxide, ozone
Weather (5)	temperature_2m, relative_humidity_2m, surface_pressure, wind_speed_10m, precipitation
Temporal (7)	hour, day_of_week, month, hour_sin, hour_cos, month_sin, month_cos
Lag (1)	aqi_lag_1 (previous hour's AQI)

Cyclical encoding for circular features:

$$\text{hour\_sin} = \sin\left(\frac{2\pi \times \text{hour}}{24}\right), \quad \text{hour\_cos} = \cos\left(\frac{2\pi \times \text{hour}}{24}\right) \tag{2.3}$$

2.4 Model Development

2.4.1 Train-Test Split

- **Method:** Temporal split (80% train, 20% test)
- **No shuffling:** Preserves time-series order
- **No data leakage:** Only past information used

2.4.2 Models Evaluated

1. **Random Forest:** Ensemble of decision trees
2. **XGBoost:** Gradient boosting with regularization
3. **LightGBM:** Histogram-based gradient boosting

2.4.3 Evaluation Metrics

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}, \quad MAE = \frac{1}{n}\sum|y_i - \hat{y}_i|, \quad RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \tag{2.4}$$

2.5 Production Pipeline

2.5.1 System Architecture

The system consists of three automated pipelines:

Table 2.3: Automation Schedule

Pipeline	**Frequency**	**Purpose**
Data Collection	Hourly	Fetch and store new data
Model Training	Daily (2 AM UTC)	Train and select best model
Prediction	Daily (3 AM UTC)	Generate 72-hour forecast

2.5.2 MongoDB Collections

1. **aqi_features:** Engineered features (updated hourly)
2. **model_registry:** Model metadata and performance
3. **predictions:** 72-hour AQI forecasts

Chapter 3

Implementation

3.1 Technology Stack

Table 3.1: Technologies Used

Component	Technology
Language	Python 3.10+
Data Processing	pandas, numpy
ML Libraries	scikit-learn, XGBoost, LightGBM
Database	MongoDB Atlas
Automation	GitHub Actions
Dashboard	Streamlit
API	OpenMeteo (air quality & weather)

3.2 Key Code Components

3.2.1 Data Collection

Listing 3.1: Fetch Air Quality Data

```
def fetch_air_quality_data(lat, lon, days=60):
    url = "https://air-quality.open-meteo.com/v1/air-quality"
    params = {
        "latitude": lat,
        "longitude": lon,
        "hourly": ["pm10", "pm2_5", "carbon_monoxide",
                   "nitrogen_dioxide", "sulphur_dioxide", "ozone"],
        "timezone": "Asia/Karachi",
        "past_days": days
    }
    response = requests.get(url, params=params)
    return pd.DataFrame(response.json()['hourly'])
```

3.2.2 AQI Calculation

Listing 3.2: Calculate Overall AQI

```python
def calculate_overall_aqi(row):
    aqi_values = [
        calculate_aqi('pm2_5', row['pm2_5']),
        calculate_aqi('pm10', row['pm10']),
        calculate_aqi('ozone', row['ozone']),
        calculate_aqi('no2', row['nitrogen_dioxide']),
        calculate_aqi('so2', row['sulphur_dioxide']),
        calculate_aqi('co', row['carbon_monoxide'])
    ]
    return max([v for v in aqi_values if v is not None])
```

3.2.3 Model Training

Listing 3.3: Train LightGBM Model

```python
from lightgbm import LGBMRegressor

model = LGBMRegressor(
    learning_rate=0.05,
    num_leaves=50,
    max_depth=20,
    n_estimators=200,
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

3.3 GitHub Actions Workflow

Listing 3.4: Hourly Data Collection Workflow

```yaml
name: Hourly Data Collection
on:
  schedule:
    - cron: '0 * * * *'
jobs:
  collect-data:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-python@v4
        with:
          python-version: '3.10'
      - run: pip install -r requirements.txt
      - run: python src/pipeline/collect_and_store_features.py
```

Chapter 4

Results and Analysis

4.1 Model Performance

Table 4.1: Model Comparison Results

Model	**R^2**	**MAE**	**RMSE**
Random Forest	0.7845	16.34	38.21
XGBoost	0.8124	15.12	35.67
LightGBM	**0.8229**	**14.68**	**34.86**

4.1.1 Best Model: LightGBM

Selected for production based on:

- Highest R^2 (0.8229) - explains 82.29% of AQI variance
- Lowest MAE (14.68) - average error 15 AQI units
- Lowest RMSE (34.86) - best handling of large errors
- Faster training and lower memory footprint

4.2 Performance Interpretation

4.2.1 Why $R^2 = 0.82$ is Excellent

This performance is realistic and production-ready because:

1. **Inherent Uncertainty:** Air quality is influenced by unpredictable factors (traffic, weather, emissions)
2. **Proper Validation:** Time-series split prevents data leakage
3. **No Future Information:** Only past data used for prediction
4. **Honest Metrics:** No overfitting or inflated performance

Models with $R^2 > 0.95$ often indicate data leakage or overfitting.

4.2.2 Error Context

Table 4.2: MAE by AQI Category

Category	AQI Range	MAE
Good	0-50	8.34
Moderate	51-100	12.56
Unhealthy (Sensitive)	101-150	18.92
Unhealthy	151-200	24.71

Average error (14.68) is within the same or adjacent AQI category, making predictions reliable for health advisories.

4.3 Feature Importance

Table 4.3: Top 10 Important Features

Rank	Feature	Importance
1	aqi_lag_1	28.5%
2	pm2_5	19.8%
3	pm10	15.6%
4	ozone	8.9%
5	hour	6.7%
6	temperature_2m	5.4%
7	relative_humidity_2m	4.2%
8	nitrogen_dioxide	3.8%
9	month	2.9%
10	wind_speed_10m	2.3%

Key Insights:

- Previous AQI (28.5%) is most influential
- Particulate matter (PM2.5 + PM10) accounts for 35.4%
- Temporal features (hour + month) contribute 9.6%

Chapter 5

Conclusions and Future Work

5.1 Key Achievements

1. **Complete ML Pipeline:** Built end-to-end system from data collection to deployment
2. **Strong Performance:** Achieved $R^2 = 0.8229$ with realistic validation
3. **Full Automation:** Hourly data collection, daily training and prediction
4. **Production Ready:** MongoDB integration, Streamlit dashboard, 99.4% uptime

5.2 Technical Learnings

- API evaluation and selection strategies
- Time-series validation techniques
- Feature engineering for environmental data
- Production ML system design
- Cloud deployment (MongoDB Atlas, GitHub Actions)
- MLOps practices (versioning, monitoring, automation)

5.3 Challenges Overcome

Table 5.1: Key Challenges and Solutions

Challenge	**Solution**
DNS blocking of AQICN/OpenWeather	Evaluated multiple APIs, selected OpenMeteo
Limited historical data (2 months)	Continuous collection strategy, 60+ days initial
Proper time-series validation	Implemented temporal split, no shuffling
Production reliability	Error handling, retry logic, monitoring

5.4 Limitations

1. Single location (Islamabad only)
2. API data has 1-2 hour delay
3. Performance degrades for extreme AQI values (limited training data)
4. No confidence intervals (point predictions only)

5.5 Impact

This project demonstrates that sophisticated environmental monitoring systems can be built using open-source tools and public data. Key impacts:

- **Public Health:** Provides Islamabad residents with reliable 3-day AQI forecasts
- **Accessibility:** Open methodology and free data sources
- **Transparency:** Complete documentation of methods and performance
- **Education:** End-to-end ML project example

5.6 Final Remarks

This internship at 10Pearls provided invaluable experience in building production ML systems. The AQI Predictor encompasses the complete machine learning lifecycle—from data collection and feature engineering to deployment and monitoring—using industry-standard tools and best practices.

The system achieves strong predictive performance ($R^2 = 0.82$) with realistic validation, operates reliably (99.4% uptime), and provides actionable forecasts for public health decision-making. While opportunities for enhancement exist, the current system demonstrates that open, transparent, and effective environmental monitoring is achievable with accessible technology.

Appendix A

Project Information

A.1 GitHub Repository

https://github.com/u-faizan/Pearls-AQI-Predictor

A.2 Directory Structure

```
AQI_Predictor/
 src/
    data/data_collector.py
    features/calculate_aqi.py
    features/feature_engineering.py
    pipeline/
        collect_and_store_features.py
        train_and_register_model.py
        predict_next_3_days.py
 .github/workflows/
    hourly_data_collection.yml
    daily_model_training.yml
    daily_prediction.yml
 dashboard/streamlit_app.py
 requirements.txt
 README.md
```

A.3 Key Dependencies

```
pandas==2.1.4
numpy==1.26.2
scikit-learn==1.3.2
xgboost==2.0.3
lightgbm==4.1.0
pymongo==4.6.1
streamlit==1.29.0
requests==2.31.0
```

Appendix B

EPA AQI Standards

B.1 AQI Categories

Table B.1: EPA AQI Categories

Range	**Category**	**Health Implications**
0-50	Good	Air quality is satisfactory
51-100	Moderate	Acceptable for most people
101-150	Unhealthy (Sensitive)	May affect sensitive groups
151-200	Unhealthy	Everyone may experience effects
201-300	Very Unhealthy	Health alert
301+	Hazardous	Emergency conditions

B.2 PM2.5 Breakpoints

Table B.2: PM2.5 AQI Breakpoints (µg/m^3)

AQI Range	**PM2.5 Low**	**PM2.5 High**
0-50	0.0	12.0
51-100	12.1	35.4
101-150	35.5	55.4
151-200	55.5	150.4
201-300	150.5	250.4
301-500	250.5	500.4

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